HH-B-681b
January 3, 1964

SUPERSEDING
Int. Fed. Spec. HH-B-00681a (GSA-FSS)
January 15, 1963 and
Fed. Spec. HH-B-681
May 2, 1933

FEDERAL SPECIFICATION

BRICK, REFRACTORY, SILICA

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one grade of silica brick for use in steel and glass furnaces.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-640—Boxes, Fiberboard, Corrugated, Triple Wall.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-L-10547—Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to

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the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials (ASTM) Standards:

C 16-49—Method of Testing Refractory Brick Under Load at High Temperatures.
C 18-60—Methods for Chemical Analysis of Refractory Materials.
C 20-46—Test for Apparent Porosity, Water Absorption, Apparent Specific Gravity, and Bulk Density of Burned Refractory Brick.
C 133-55 — Test for Cold Crushing Strength and Modulus of Rupture of Refractory Brick and Shapes.
C 134-41—Methods of Test for Size and Bulk Density of Refractory Brick.
C 154-41—Method of Test for Warpage of Refractory Brick and Tile.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Material. Silica brick shall be composed of ground silica rock (quartzite or ganister) and a suitable binder that has been burned or fired to produce the desired strength and structure.

3.2 Composition. When analyzed in accordance with 4.4.1, bricks shall contain the chemical compositions shown in table I.

TABLE I—*Chemical composition*

Element	Percentage	
	Minimum	Maximum
Silica (SiO_2)	95.	—
Alumina (Al_2O_3)	—	1.00
Iron Oxide (Fe_2O_3)	—	0.90
Titania (TiO_2)	—	0.10
Lime (CaO)	—	3.00
Alkalies (Na_2O, K_2O) ..	—	0.10
Magnesia (MgO)	—	0.12

3.3 Dimensions. Silica brick shall be furnished in the sizes, shapes, and dimensions specified (see 6.2).

3.4 Tolerances. The dimensions specified for silica brick, sized 4 inches and larger, shall not vary as a result of shrinkage by more than plus or minus 2 percent. When dimensions specified are less than 4 inches, a tolerance of plus or minus three percent shall be allowed for shrinkage (see 4.4.2).

3.4.1 *Warpage.* When tested in accordance with paragraph 4.4.3, the tolerance for the warpage of silica brick shall show no warpage greater than 1 percent of the diagonal measurement.

3.5 Physical properties.

3.5.1 *Apparent specific gravity.* When tested in accordance with 4.4.4, the apparent specific gravity of silica brick shall be more than 2.30.

3.5.2 *Modulus of rupture.* When tested in accordance with 4.4.5, the modulus of rupture shall average not less than 600 pounds per square inch.

3.5.3 *Apparent porosity.* Bricks shall have an apparent porosity of 27 percent maximum for a standard 9-inch brick when determined in accordance with 4.4.6.

3.5.4 *Load test at high temperatures.* When tested in accordance with 4.4.7, bricks shall have a minimum shear load test failure temperature of 3000°F.

3.6 Marking. The trade name or the name of the manufacturer shall be molded into each brick.

3.7 Workmanship. The quality of workmanship shall be such as to produce bricks that are compact, free of voids, cracks, checkers, soft centers, and excessive warpage, in accordance with the requirements of this specification.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified (see 6.2), the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 *Inspection lot.* All brick of the same shape and size not exceeding 10,000 offered for delivery at one time shall be considered a lot for purposes of inspection.

4.2.2 *Sampling for examination.* A random sample of brick shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level II. The acceptable quality levels shall be 1.5 percent defective units for major defects and 4.0 percent defective units for minor defects.

4.2.3 *Sampling for tests.* A random sample of brick shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level L8.

4.3 Examination. Samples selected in accordance with 4.2.2 shall be examined for the defects listed in table II. Lot acceptance or rejection shall be based on an acceptable quality level (AQL) of 2.5 percent defective for major defects and 4.0 percent defective for minor defects. Examination shall be conducted as specified in table II.

TABLE II—*Classification of defects in accordance with MIL-STD-105*

Categories	Defects
Major:	
101	Dimensions not as specified and not within tolerance.
102	Surface cracks.
103	Void spaces.
104	Excessive warpage.
105	Checkers.
Minor:	
201	Marking not correct.

4.3.1 *Inspection of preparation for delivery.* The packing and marking of the bricks shall be inspected to determine conformance with the applicable requirements of section 5 of this specification. The packing and marking requirements of section 5 identified to referenced specifications and standards shall be inspected in accordance with applicable referenced specifications and standards. The packing requirements of section 5 not identified to referenced specification shall be examined for defects listed in 4.3.1.1 in accordance with MIL-STD-105 based on an AQL of 4.0 percent defective.

4.3.1.1 *Classification of defects.* Packing defects shall be classified in accordance with table III.

TABLE III—*Classification of defects for packing*

Categories	Defects
Major:	
101	Cushioning not provided.
102	Boxes not modified with addition of skids as specified.
103	Case liners not provided.
104	Closure and strapping not as specified.
105	Strapping not zinc-coated when applicable.
106	Pallets not provided when specified.

4.4 Test procedures.

4.4.1 *Chemical analysis.* Chemical analysis shall be determined in accordance with ASTM Method C 18-60.

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4.4.2 *Size.* The size of bricks, sampled in accordance with 4.2.3, shall be determined as specified in ASTM Method C 134-41.

4.4.3 *Warpage.* The warpage of bricks, sampled in accordance with 4.2.3, shall be determined as specified in ASTM Method C 154-41.

4.4.4 *Apparent specific gravity.* The apparent specific gravity of bricks shall be determined in accordance with ASTM Method C 20-46.

4.4.5 *Modulus of rupture.* The modulus of rupture of bricks shall be determined in accordance with ASTM Method C 133-55.

4.4.6 *Apparent porosity.* The apparent porosity of bricks shall be determined in accordance with ASTM Method C 20-46.

4.4.7 *Load test.* The resistance to shear loading shall be determined in accordance with schedule 4 of ASTM Method C 16-49.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and application of the levels for packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* The bricks shall be packed in a nailed and lock-corner wood box or cleated plywood box conforming to PPP-B-621 and PPP-B-601, respectively. Cushioning shall be used to fill all voids to insure a compact fit. Boxes exceeding 250 pounds gross weight shall be modified by the addition of not less than two nominal 2-inch by 4-inch wood skids attached with nails driven from the inside of the box, flush with the outside edges. Each of the skids shall be beveled full depth at a 45 degree angle. Shipping containers shall be lined with sealed case liners conforming to MIL-L-10547 and the appendix thereto. The cleated plywood boxes shall be closed and strapped in accordance with the appendix thereto. Flat steel strapping shall be type I, class B.

5.1.2 *Level B.* The bricks shall be packed in a class 2, grade A, style G triple-wall corrugated fiberboard box with pallet base conforming to PPP-B-640. Cushioning shall be used to fill all voids to insure compact fit. Pack shall be strapped with two steel straps lengthwise and three steel straps girthwise spaced equidistant.

5.1.3 *Level C.* The bricks shall be packed in a manner that will be acceptable to the carrier at lowest rates and will insure arrival at destination in a satisfactory condition.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special markings required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. Silica bricks covered by this specification are intended for use in steel and glass furnaces. Silica bricks are not intended for the lining of fire boxes of boilers.

6.2 Ordering data. Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Sizes and dimensions required (see 3.3).
(*c*) Responsibility for inspection (see 4.1).
(*d*) Level of packing required (see 5.1).

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6.3 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Bricks, fire, not otherwise indexed, including fire brick shapes.
Carload minimum weight 60,000 pounds.

Motor:

Bricks, fire, including fire brick shapes.
Truckload minimum weight 60,000 pounds, subject to Rule 115, National Motor Freight Classification.

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MILITARY CUSTODIANS:

Army—MO

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Copies of this specification may be purchased for 5 cents each.

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